VERITAS UNIVERSITY, ABUJA · DEPARTMENT OF SOFTWARE ENGINEERING

VUA-SEN307 Database Systems — Week 4 Practical

Lab 2: Database Design and Normalization

Duration: 3 hours **Tools:** MySQL Workbench (EER Diagram tool), draw.io (optional), Git

Platforms: Windows & macOS

Builds on: your Week 3 setup (Level 1 Connection and your `sen307_<matric_number>` database)

Objectives

By the end of today you should be able to:

- turn a real-world scenario into an Entity-Relationship (ER) diagram
- identify relationships and cardinalities (one-to-many, many-to-many)
- normalize a table step by step from 1NF to 2NF to 3NF to remove redundancy

Session plan

Time	Activity
0:00 – 0:15	Recap and setup check: open the Level 1 Connection
0:15 – 0:45	**Part 1:** Identify entities, attributes and relationships
0:45 – 1:30	**Part 2:** Normalize the registration table (1NF to 3NF)
1:30 – 1:40	Break
1:40 – 2:30	**Part 3:** Build the EER diagram in MySQL Workbench
2:30 – 2:50	**Part 4:** Write-up and GitHub commit
2:50 – 3:00	Submission check and wrap-up

Windows & macOS: In MySQL Workbench the steps are the same on both. Where a step differs, it is noted in that step.

Recap and Setup Check (15 min)

1. Open MySQL Workbench and click the **Level 1 Connection** tile.
2. Run the commands below, replacing `20231234` with your matric number:

```
SHOW DATABASES;
USE sen307_20231234;
SELECT DATABASE();
```

3. If your connection or database is missing, tell the Lab Technician now. You need a working setup before Week 5.

Part 1: Entities, Attributes and Relationships (30 min)

Scenario: Design a database for a university course-registration system covering **Students, Courses, Lecturers, Departments and Enrolments**.

- A student registers for many courses in a semester.
- A course may have multiple lecturers across different sessions.

Step 1: List the entities and their attributes

Copy this table into your notes and complete it:

Entity	Attributes	Primary key
Department		
Student		
Lecturer		
Course		
Enrolment		

Step 2: Identify the relationships and their cardinalities

Answer these:

- How many departments does one student belong to? How many students can one department have?
- How many courses can one student register for? How many students can one course have?
- How many lecturers can one course have, across sessions? How many courses can one lecturer teach?

> **Hint:** A **many-to-many (M:N)** relationship can't be stored directly in a relational database. You need a **linking (junction) table** between the two entities. Check whether your design has two M:N relationships.

Part 2: Normalization from 1NF to 3NF (45 min)

Work on paper or in a document. You are given this **unnormalized** registration sheet:

matric_no	student_name	student_dept	course_codes	course_titles	lecturers	lecturer_depts	session
20231234	Ada Obi	Software Engineering	SEN307, SEN301	Database Systems, Software Architecture	Dr A. Okafor, Dr M. Bello	Software Eng., Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307, COS301	Database Systems, Operating Systems	Dr A. Okafor, Mrs F. Eze	Software Eng., Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

For each stage, show the resulting table(s) and write **one sentence** explaining what you fixed.

Step 1: First Normal Form (1NF)

- Remove multi-valued cells, so that each cell holds **one value only**.
- Give each student–course pair its own row.
- Choose a primary key for the new table. *(Hint: no single column is unique on its own.)*

Step 2: Second Normal Form (2NF)

- Find attributes that depend on only **part** of the composite key. These are partial dependencies.
  - For example: does `student_name` depend on the course, or only on `matric_no`?
- Move each group into its own table, so you end up with Student, Course and Enrolment.

Step 3: Third Normal Form (3NF)

- Find **transitive dependencies**, where a non-key column depends on another non-key column.
  - For example: `student_dept` describes a department, not the student. What happens if a department is renamed?
- Move the department details into a Department table and link to it with a foreign key.
- Handle lecturers the same way, and decide where to record *which lecturer teaches which course in which session*.

Step 4: Write out your final schema

Use this format:

```
Department (dept_id PK, dept_name)
Student    (student_id PK, matric_no, full_name, dept_id FK → Department)
...
```

Part 3: The EER Diagram in MySQL Workbench (50 min)

1. Go to **File → New Model**. On Windows the shortcut is **Ctrl + N**; on macOS it is **Cmd + N**.
2. Under **Physical Schemas**, double-click `mydb` and rename it to `sen307_<matric_number>`.
3. Double-click **Add Diagram**. A blank EER canvas opens.
4. **Add the tables.** Click the **Place a New Table** tool (the table icon in the left toolbar), then click on the canvas. Double-click the new table to open its editor below the canvas, and set:
   - the **table name**
   - the **columns and data types**, for example `INT`, `VARCHAR(100)` or `DATE`
   - **PK** and **NN** (Not Null) on the primary key, plus **AI** (Auto Increment) for `INT` IDs
5. Repeat step 4 for every table in your 3NF schema.
6. **Add the relationships** using the relationship tools in the left toolbar:
   - Use **1:n Non-Identifying Relationship** for links such as Department → Student. To draw it, click the *child* table (Student) first, then the *parent* table (Department).
   - Workbench adds the foreign-key column for you. Rename it to match your schema if needed, for example `dept_id`.
7. Arrange the tables neatly so that no lines cross where you can avoid it. Make sure every **PK** (key icon) and **FK** (red diamond) is visible.
8. Save the model as `lab2_model.mwb`.
9. Export the diagram with **File → Export → Export as PNG**, and save it as `lab2_erd.png`.

> **macOS:** If the export menu is greyed out, click once on the diagram canvas first.
> **draw.io (optional):** You may sketch a first draft at `app.diagrams.net` . The **final** diagram must still come from MySQL Workbench.

This exported PNG is Submission 1.

Part 4: Write-up and GitHub (20 min)

1. Write a **one-page** document, `lab2_normalization.pdf` or `.md`, containing:
   - the unnormalized table
   - your 1NF, 2NF and 3NF tables, each with a one-sentence explanation
   - your final schema with PKs and FKs marked
2. Commit your work to your repository:

```
cd sen307-labs
mkdir week04
```

3. Copy `lab2_model.mwb`, `lab2_erd.png` and your write-up into `week04/`. On Windows, use File Explorer; on macOS, use Finder.
4. Commit and push:

```
git add .
git commit -m "Week 4: ER diagram and normalization to 3NF"
git push
```

Submission Checklist (before you leave)

- ☐ ER diagram exported as PNG or PDF from MySQL Workbench (`week04/lab2_erd.png`)
- ☐ One-page normalization write-up showing 1NF, 2NF and 3NF (`week04/lab2_normalization`)
- ☐ Workbench model file (`week04/lab2_model.mwb`)
- ☐ All files pushed to GitHub

Next week (Week 5): you'll turn this schema into real tables with `CREATE TABLE`, so keep your `.mwb` model file safe.